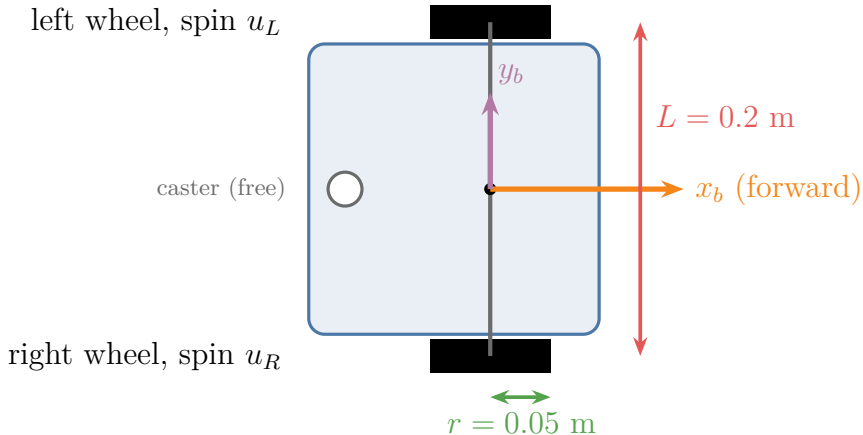




midpoint of the axle = origin of the body frame